

Number: P3367R4
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Audience: CWG, LWG
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Target: C++29

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constexpr coroutines

This paper is proposing making coroutines functional during constant evaluation. Even when most of use-cases for coroutines are based on I/O and event based, coroutines are still useful for compile-time based computation, eg. `std::generator`.

Changes

- [R3 → R4](#): [allow.....co_await.....and co_yield.....only.....within.....coroutine evaluation](#)
- [R2 → R3](#): [added motivation text, added AST transform approach examples](#)
- [R1 → R2](#): [Updated wording to elide allocator.....functionality.....in.....coroutine promise types, Added discussion about alternative.....implementation.....strategies, Added.....information.....about.....decision.....by various groups which seen this paper,](#)
- [R0 → R1](#): [Changed wording mentioning lifetime of coroutines after expr.const 5.2](#)

Timeline

EWG in Wroclaw

Poll: P3367 constexpr coroutines send to CWG and LEWG for inclusion for C++26

SF	F	N	A	SA
4	12	3	0	0

Result: Consensus

LEWG in Wroclaw

Poll: Forward P3367R1 to LWG for C++26

SF	F	N	A	SA
12	5	0	0	0

Result: Unanimous consent